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# Microeconomics I

Class 5 — Indirect utility, wealth and price effects

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## Plan for today

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Warm-up; rice in Hunan

Indirect utility

Roy's identity

Break

Wealth effects: normal and inferior goods

Price effects: Giffen goods, substitutes, complements

In pairs: from  $v$  to demand; summary

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## Practicalities

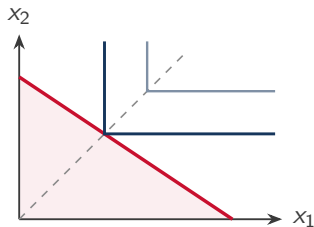
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- Problem set 1: due at the start of class tomorrow, Wed 14 Oct, on paper, one copy per group
- Office hours: Wednesdays 14:00–15:00, Room 445  
up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: [pkocourek.com/teaching/micro-1](http://pkocourek.com/teaching/micro-1) · [pkocourek@ub.edu](mailto:pkocourek@ub.edu)

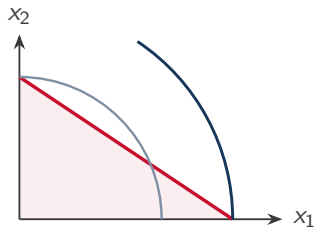
## Warm-up: where does each consumer buy?

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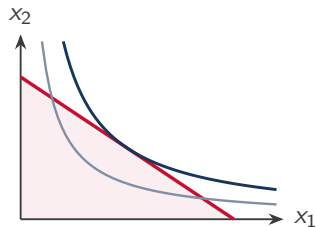
$p = (2, 3)$ ,  $w = 12$  in every panel



**L-shaped, kinks on  $x_2 = x_1$**



**bowed out**



**$x_1 x_2$  or  $\ln x_1 + \ln x_2$**

## Rice gets cheaper. Do poor households eat more rice?

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**Hunan, China, 2006:** 650 poor urban households

rice: 64 % of their calories

at random, vouchers: up to a quarter  
off the price of rice, for five months

## More rice, the same, or less?

A field experiment; what the households did, in class.

# Revealed in class

Before class: decide which answer you expect, and why.

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# Indirect utility

MWG 3.D, 3.G

## Indirect utility: the level of the best curve the budget reaches

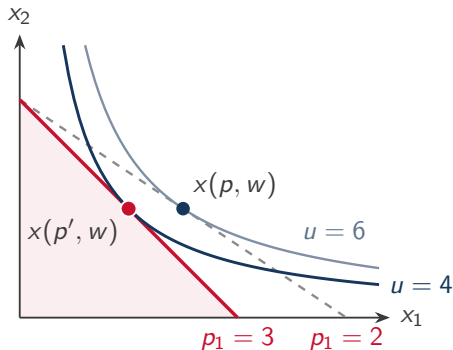
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$$v(p, w) = u(x^*) \text{ for any } x^* \in x(p, w)$$

the value of the UMP:  $\max\{u(x) : x \in \mathcal{B}_{p,w}\}$

$$u = x_1 x_2: \quad v(p, w) = \frac{w}{2p_1} \cdot \frac{w}{2p_2} = \frac{w^2}{4p_1 p_2}$$

at  $p = (2, 3)$ :  $w^2/24$  (class 4, slide 26)



$p_1: 2 \rightarrow 3$ :  $v$  falls from 6 to 4.

MWG 3.D, p. 56.



$v(p, w)$  is homogeneous, rises with  $w$  and never rises with prices

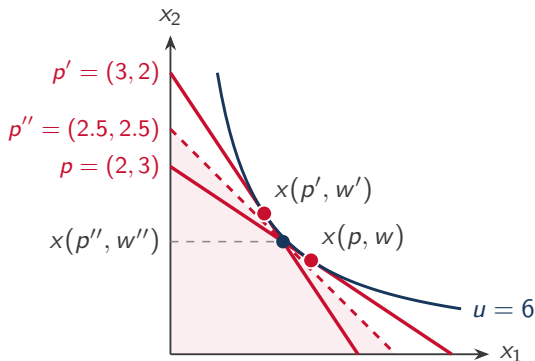
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| property                        | why                    | for $v = \frac{w^2}{4p_1p_2}$                                  |
|---------------------------------|------------------------|----------------------------------------------------------------|
| (i) homogeneous of degree 0     | the same budget set    | $\frac{(\alpha w)^2}{4 \alpha p_1 \alpha p_2} = v$             |
| (ii) strictly increasing in $w$ | LNS (class 2)          | $\frac{\partial v}{\partial w} = \frac{w}{2p_1p_2} > 0$        |
| nonincreasing in each $p_\ell$  | the budget set shrinks | $\frac{\partial v}{\partial p_1} = -\frac{w^2}{4p_1^2p_2} < 0$ |
| (iii) quasiconvex in $(p, w)$   | next slide             |                                                                |
| (iv) continuous                 |                        |                                                                |

MWG Proposition 3.D.3, p. 56:  $u$  continuous,  $\succsim$  locally non-satiated,  $X = \mathbb{R}_+^2$ .

## $v$ is quasiconvex: the average of two budgets never beats both

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$\{(p, w) : v(p, w) \leq \bar{v}\}$  is convex  
for every  $\bar{v}$

$p = (2, 3)$  or  $p' = (3, 2)$ ,  $w = 12$ :  
 $v = 6$  both

their average  $(2.5, 2.5)$ ,  $w = 12$ :  
 $v = 5.76$

What the average budget affords, one of the two already affords:  $v \leq 6$ .

MWG Proposition 3.D.3(iii), p. 56; Figure 3.D.5, p. 57, in Wednesday's reading;  $u = x_1 x_2$ .

## In pairs: could this be an indirect utility function?

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Test (i) and (ii) of Prop. 3.D.3 (slide 9). Four minutes.

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|             |                            |                                         |
|-------------|----------------------------|-----------------------------------------|
| $v(p, w) =$ | homogeneous<br>of degree 0 | $\uparrow$ in $w$ , $\downarrow$ in $p$ |
|-------------|----------------------------|-----------------------------------------|

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$$\frac{w}{p_1 + p_2}$$

$$\frac{w^2}{p_1 p_2}$$

$$\frac{p_1 + p_2}{w}$$

$$\ln w - \ln p_1 - \ln p_2$$

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Answers revealed in class. Try it in pairs before class.

## Change the utility function: $v$ and $\lambda$ change, demand does not

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| $u$                                         | $v(p, w)$                                                   | $\lambda = \partial v / \partial w$<br>at $p = (2, 3), w = 12$ |
|---------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------|
| $x_1 x_2$                                   | $\frac{w^2}{4p_1 p_2}$                                      | $\frac{w}{2p_1 p_2} = 1$                                       |
| $\frac{1}{2} \ln x_1 + \frac{1}{2} \ln x_2$ | $\ln w - \frac{1}{2} \ln p_1 - \frac{1}{2} \ln p_2 - \ln 2$ | $\frac{1}{w} = \frac{1}{12}$                                   |

Same  $\succsim$ , same  $x(p, w) = (3, 2)$ ;  $v$  becomes  $f(v)$ , and  $\lambda$  with it.

MWG p. 57, in Wednesday's reading ( $\tilde{u} = f(u)$  gives  $\tilde{v} = f(v)$ ; Example 3.D.2 with  $\alpha = \frac{1}{2}$ );  
 $\lambda = \partial v / \partial w$ : pp. 54–55, class 4, slide 26.

## Roy's identity: demand from two slopes of $v$

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$$x_\ell(p, w) = - \frac{\partial v(p, w) / \partial p_\ell}{\partial v(p, w) / \partial w}$$

$u$  continuous;  $\succsim$  locally non-satiated and strictly convex;  $v$  differentiable at  $(p, w) \gg 0$

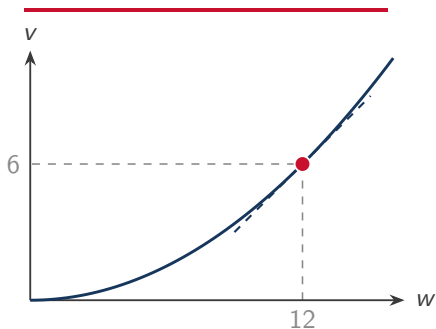
$$v = \frac{w^2}{4p_1p_2}: \quad - \frac{-w^2/(4p_1^2p_2)}{w/(2p_1p_2)} = \frac{w}{2p_1}$$

Cobb–Douglas, class 4

Replace  $u$  by  $f(u)$ : both slopes gain the factor  $f'$ , and the ratio does not change.

MWG Proposition 3.G.4 (Roy's identity), p. 73.

## A dearer good costs utility in proportion to how much of it she buys

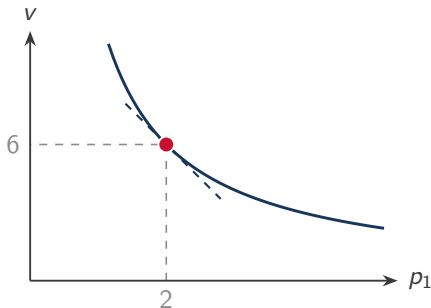


$p = (2, 3)$ : slope  $\lambda = 1$

$$\frac{\partial v}{\partial w} = \lambda, \quad \frac{\partial v}{\partial p_\ell} = -\lambda x_\ell(p, w)$$

Here  $x_1 = -(-3)/1 = 3$ : the bundle  $(3, 2)$  again.

MWG p. 74, Proofs 2 and 3 (the envelope theorem);  $u = x_1 x_2$ ,  $p = (2, 3)$ ,  $w = 12$ .



$w = 12$ ,  $p_2 = 3$ : slope  $-\lambda x_1 = -3$

## (optional) A bliss point: does more wealth always raise $v$ ?

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$$u(x) = -(x_1 - 2)^2 - (x_2 - 2)^2 \text{ on } \mathbb{R}_+^2, \quad p = (2, 3).$$

(1)  $v(p, w)$  for  $w \geq 10$ .

(2) Which part of Prop. 3.D.3 fails?

Why?

(3)  $\lambda = \partial v / \partial w$  at  $w = 12$ ?

At home if not done in class; the answers are in the slides released after class.

MWG Proposition 3.D.3, p. 56 (slide 9).

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# Break

Ten minutes



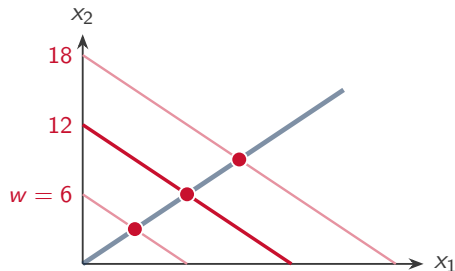
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# Wealth and price effects

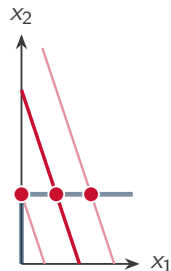
MWG 2.E

## More wealth: Cobb–Douglas scales up; quasilinear: all to good 1

Engel function:  $w \mapsto x(p, w)$ ,  $p$  fixed; wealth effect  $\partial x_\ell / \partial w$



$$u = x_1 x_2, \quad p = (2, 3)$$
$$x = (w/4, w/6): \text{a ray}$$

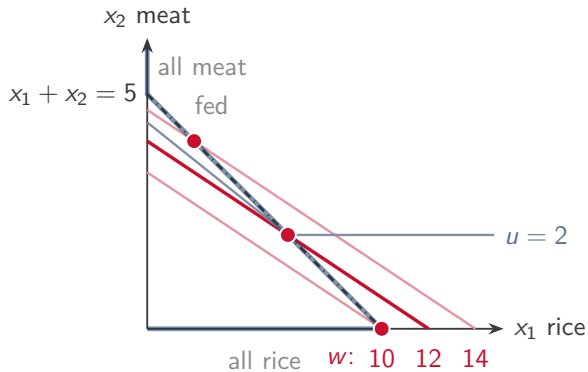


$$u = x_1 + \ln(1 + x_2), \quad p = (3, 1)$$
$$x_2 = 2 \text{ once } w \geq 2$$

Class 4's promise: past  $w = 2$ , every extra euro goes to good 1.

MWG 2.E, pp. 24–25, Figure 2.E.1 (the wealth expansion path).

## Rice can be inferior: past subsistence, more wealth buys less of it



$$p = (2, 3)$$

she needs 5 units of food:

$$x_1 + x_2 \geq 5$$

once fed, only meat counts

normal at  $(p, w)$ :  $\partial x_\ell / \partial w \geq 0$ ;

inferior:  $< 0$

Rice is normal up to  $w = 10$ , inferior from 10 to 15.

$u = \min\{x_2, x_2 - 4(5 - x_1 - x_2)\}$ : meat, minus 4 per unit of missing food. MWG 2.E, p. 25; own example.

## The budget ties the effects together: not every good can be inferior

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$$\sum_k p_k \frac{\partial x_\ell}{\partial p_k} + w \frac{\partial x_\ell}{\partial w} = 0$$

homogeneity: Prop. 2.E.1

$$\sum_\ell p_\ell \frac{\partial x_\ell}{\partial w} = 1$$

Walras in  $w$ : Engel aggregation, Prop. 2.E.3

$$\sum_\ell p_\ell \frac{\partial x_\ell}{\partial p_k} + x_k = 0$$

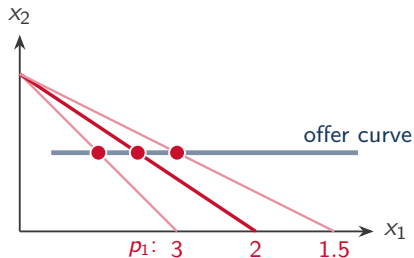
Walras in  $p_k$ : Cournot aggregation, Prop. 2.E.2

Slide 19 at  $w = 12$ :  $2 \cdot (-1) + 3 \cdot (+1) = 1$ . An extra euro is spent somewhere.

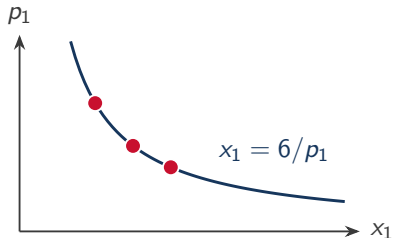
MWG 2.E, pp. 27–28.

## Cheaper good 1: Cobb–Douglas buys more of it, as much of good 2

usual:  $\partial x_\ell / \partial p_\ell \leq 0$ ; Giffen at  $(p, w)$ :  $\partial x_\ell / \partial p_\ell > 0$



**goods: the offer curve**



**prices: the demand curve**

$x_1 = 6/p_1$  rises as  $p_1$  falls;  $x_2$  stays at 2: the offer curve is flat.

MWG 2.E, pp. 25–26, Figures 2.E.2–2.E.3;  $u = x_1 x_2$ ,  $w = 12$ ,  $p_2 = 3$ .

## Revealed in class

Before class: on slide 19's picture, let rice cost 1.5 instead of 2  
and find her new best bundle.

## Dearer good 2: $x_1$ falls, stays or rises, depending on $\sigma$

gross substitutes:  $\partial x_1 / \partial p_2 > 0$ ; gross complements:  $< 0$

| $\sigma$ | $u$                       | $x_1(p, w)$                    | $p_2: 3 \rightarrow 4$<br>$p_1 = 2, w = 12$ |
|----------|---------------------------|--------------------------------|---------------------------------------------|
| 0        | $\min\{x_1, x_2\}$        | $\frac{w}{p_1 + p_2}$          | 2.4 $\rightarrow$ 2: complements            |
| 1        | $x_1 x_2$                 | $\frac{w}{2p_1}$               | 3 $\rightarrow$ 3: neither                  |
| 2        | CES, $\rho = \frac{1}{2}$ | $\frac{w p_2}{p_1(p_1 + p_2)}$ | 3.6 $\rightarrow$ 4: substitutes            |

CES: complements for  $\sigma < 1$ , neither at  $\sigma = 1$ , substitutes for  $\sigma > 1$ .

MWG 3.G, p.70 ("gross" = Walrasian demand; the Hicksian version is Wednesday). CES as class 4, slide 29.

## In pairs: Roy's identity, then the three effects

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Find the demand by Roy's identity. Wealth effect? Own price: usual or Giffen?

The other price rises: substitutes or complements? Seven minutes.

| $v(p, w)$                                                       | demand | wealth | own price | other price $\uparrow$ |
|-----------------------------------------------------------------|--------|--------|-----------|------------------------|
| $\frac{w^3}{p_1 p_2^2}$ , find $x_1$                            |        |        |           |                        |
| $\frac{w + p_2}{p_1} - 1 + \ln \frac{p_1}{p_2}$ ,<br>find $x_2$ |        |        |           |                        |
| class 4's $x_1 + \ln(1 + x_2)$ , $p_1 > p_2$                    |        |        |           |                        |
| $\frac{w - 5p_1}{p_2 - p_1}$ , find $x_1$                       |        |        |           |                        |
| slide 22's consumer, near $p = (2, 3)$ , $w = 12$               |        |        |           |                        |

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Answers revealed in class. Try it in pairs before class.



## (optional) Good 1 is Giffen: what happens to $x_2$ when $p_1$ rises?

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A consumer with two goods; Walras' law holds;  $\partial x_1 / \partial p_1 > 0$ .

(1) Show that  $\partial x_2 / \partial p_1 < 0$ .

Hint: slide 20, line 3, with  $k = 1$ .

(2) Explain (1) with her spending.

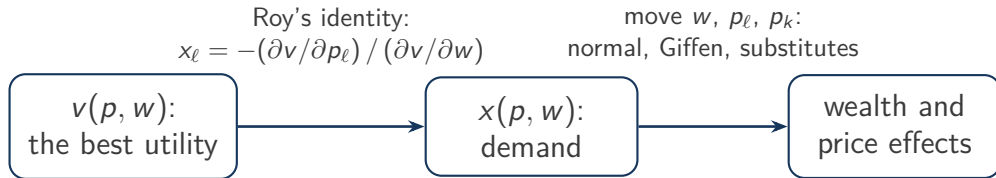
(3) Can  $\partial x_1 / \partial p_2 > 0$  hold too?

At home if not done in class; the answers are in the slides released after class.

MWG 2.E, pp. 27–28 (Proposition 2.E.2, Cournot aggregation).

## Today in one line

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Wednesday: expenditure minimisation — Hicksian demand, Shephard's lemma, duality.

Problem set 1: due at the start of class tomorrow, on paper, one copy per group.

Problem set 2 is handed out tomorrow; due in class Wednesday 21 October.

Read MWG 3.E–3.G (or Muñoz-García ch. 2) for Wednesday 14 October.